LEARN MATES

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Note from author

These notes are mostly based on the syllabus and past papers

Fair warning: they might look short (2–3 pages), but I've condensed everything down to keep the important stuff while removing the extra yap. That said, I wouldn't recommend using these as quick revision notes, but think of them as detailed notes covering almost everything

Spotted a mistake? Suggestions?
I'd love to hear your feedback. Contact us at: learnmates.share@gmail.com

The Elements of Groups 1 and 2

Properties

General

- Their color in their pure form tends to be bright silvery; however, when exposed to air they get oxidised and appear dull
- Conductive of heat and electricity
- All their simple salts (eg. KCl) are white in color

As you do down the group:

- Atomic radius increases
- Density increases
- Melting/ boiling points decreases

Reason: Weaker metallic bonding as atomic radius increases (greater distance between nucleus and delocalised electrons means weaker attraction).

- More reactive

Reason: increase in atomic radius (further distance between outermost e^- and nucleus) + more shielding → outweigh increase in nuclear charge first → Weaker attraction between outermost e^- and nucleus → less first ionization energy (and second for Group 2) → easier to remove (outermost) electrons → more reactive

(ionization energy has been explained in detail in the "Atomic Structure and the Periodic Table" notes on learnmates.org)

Reactions of group 1 and 2 metals

With Oxygen

General equations for:

Group 1: $4M_{(s)} + O_{2(g)} \rightarrow 2M_2O_{(s)}$

Group 2: $2M_{(s)} + O_{2(g)} \rightarrow 2MO_{(s)}$

With Chlorine

General equations for:

Group 1: $2M_{(s)} + Cl_{2(g)} \rightarrow 2MCl_{(s)}$

Group 2: $M_{(s)} + Cl_{2(g)} \rightarrow MCl_{2(s)}$

With water

General equations for:

Group 1: $2M_{(s)} + 2H_2O_{(l)} \rightarrow 2MOH_{(aq)} + H_{2(g)}$

Group 2: $M_{(s)} + 2H_2O_{(l)} \rightarrow M(OH)_{2(aq)} + H_{2(g)}$

exception:

1.Mg: very slow reaction with water (insoluble $Mg(OH)_2$ coating forms)

2.$Ca(OH)_2$ is only slightly soluble → forms white suspension/milky appearance

$Ca_{(s)} + 2H_2O_{(l)} \rightarrow Ca(OH)_{2(s/aq)} + H_{2(g)}$

Observations:

1.Reaction becomes more vigrous down the group (may even catch fire)

2.Tempreture increase

3.Fizzing (Hydrogen) —— Only Rxn with water

4.Metal dissolves when it reacts ——— Only Rxn with water (with Ca and Mg exceptions)

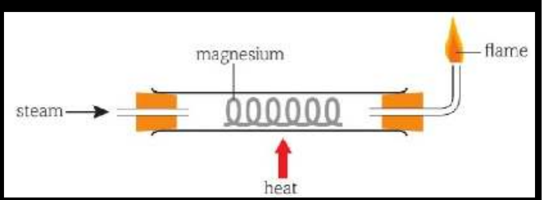
Magnesium with Steam

$Mg_{(s)} + H_2O_{(g)} \rightarrow MgO_{(s)} + H_{2(g)}$

observations:

1.Producing a bright white flame

2.White ash (MgO)



Reactions of group 1 and 2 metal oxides

With water

General equations for:

Group 1: $M_2O_{(s)} + H_2O_{(l)} \rightarrow 2MOH_{(aq)}$

Group 2: $MO_{(s)} + H_2O_{(l)} \rightarrow M(OH)_{2(aq)}$

exception:

similar to that od the reaction of Mg and Ca metals

Alkalinity:

Down group 2 the alkanlinity increases as the solubility (of hydroxides) increases:

$Mg(OH)_2$: ph 9 (sparingly soluble)

$Ca(OH)_2$: ph 12 (slightly soluble but more that $Mg(OH)_2$)

$St(OH)_2$: ph 13

$Ba(OH)_2$: ph 14

With Acids

Both hydroxides and oxides are bases so they react with acids in a neutralization reaction.

General equation:

Metal oxide/hydroxide + Acid → Salt + water

Examples:

$Na_2O_{(s)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(l)}$

$MgO_{(s)} + H_2SO_{4(aq)} \rightarrow MgSO_{4(aq)} + H_2O_{(l)}$

$Ca(OH)_{2(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + H_2O_{(l)}$

Observation:

The white solid base will dissolve to form a colorless solution.

Solubility of group 2 hydroxides and sulfates

Group 2 Hydroxides

Rule: As you go down the group solubility increases.$Mg(OH)_2$ is not (really slightly) soluble, while barium hydroxide is very soluble.

Alkalinity: increases (As mentioned before)

Group 2 Sulfates

Rule: As you go down the group solubility decreases.$MgSO_4$ is soluble, while $BaSO_4$ is not (really slightly) soluble.

Note: As for group 1 all their sulfates, hydroxides..(other compounds) are soluble

Thermal Stability of group 1 & 2 nitrates and carbonates

Thermal stability: a measure of how stable a compound is when heated.

Factors affecting thermal stability:

1.Polarizing power (more polarizing power → less thermally stable)

Reason: A cation with high polarizing power pulls electron density from the anion, weakening the bonds within the anion.

Trends:A)Group 2 have a greater charge ① than group 1 and are generaly smaller → more polarizing power → more energy is required to break bond

B)Down the group the ionic radius of ion(size) ② increases → Charge density decreases, so polarizing power decreases → anion breaks less easily (less distortion). → more energy is required to break bond

2.Complexity of anions (contains more complex anions → less thermally stable)

Trend: Complex ions like nitrates and carbonates can break down into more stable ions unlike Chlorides

The Elements of Groups 1 and 2

Stability of group 1 and 2 Carbonates

Trend: (1) Group 1 carbonates do not decompose, except lithium

(2) As you go down group 2 thermal stability increases(more heat is needed)

Equation:

Li: $Li_2CO_{3(s)} \rightarrow Li_2O_{(s)} + CO_{2(g)}$

Group 2: $MCO_{3(s)} \rightarrow MO_{(s)} + CO_{2(g)}$

Observation:

no observation (both solids are white + CO_2 is colorless)

Group 1	Result	Group 2	Result
Lithium carbonate	decomposition	Beryllium carbonate	decomposition
Sodium carbonate	no decomposition	Magnesium carbonate	decomposition
Potassium carbonate	no decomposition	Calcium carbonate	decomposition
Rubidium carbonate	no decomposition	Strontium carbonate	decomposition
Cesium carbonate	no decomposition	Barium carbonate	decomposition

Stability of group 1 and 2 Nitrates

Trend: (1) As you go down the group thermal stability increases(more heat is needed)

Equation:

Group 1 (except Li): $2MNO_{3(s)} \rightarrow 2MNO_{2(s)} + O_{2(g)}$

Li: $4LiNO_{3(s)} \rightarrow 2Li_2O_{(s)} + 4NO_{2(g)} + O_{2(g)}$

Group 2: $2M(NO_3)_{2(s)} \rightarrow 2MO_{(s)} + 4NO_{2(g)} + O_{2(g)}$

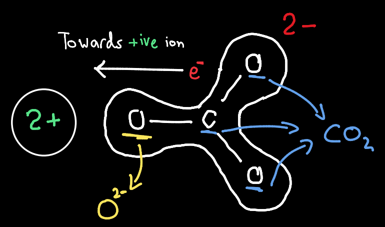
Observation:

Group 2 & Li: Acidic brown fumes (NO_2)

Group 1	Result	Group 2	Result
Lithium nitrate	brown fumes	Beryllium nitrate	brown fumes
Sodium nitrate	no brown fumes	Magnesium nitrate	brown fumes
Potassium nitrate	no brown fumes	Calcium nitrate	brown fumes
Rubidium nitrate	no brown fumes	Strontium nitrate	brown fumes
Cesium nitrate	no brown fumes	Barium nitrate	brown fumes

Explanation

The cation polarizes (distorts the electron cloud) the (large) anion, weakening the covalent bonds within the anion (e.g. the N–O bonds in nitrate or C–O bonds in carbonate), which ultimately leads to their breakdown into a smaller, more stable molecule (CO_2, NO_2) and metal oxides



Uses of group 2 metals and thier compounds

Magnesium Hydroxide (milk of magnesia):

A suspension of Magnesium Hydroxide in water. (It looks like milk, hence the name.)

Used as an Antacid (to treat indigestion/heartburn) and a mild laxative.

Why it works:

It is a base (insoluble metal hydroxide).

It neutralizes excess stomach acid (Hydrochloric acid, HCl) .

It is preferred over something like Sodium Hydroxide (NaOH) because it is insoluble → low OH^- conc. → low alkalinity , so it is safe to ingest and works slowly without damaging tissues

Lime Water [$Ca(OH)_2$]:

1–Testing for CO_2

When CO_2 is bubbled in lime water it turns cloudy as white calcium carbonate is produced:

–$Ca(OH)_{2(aq)} + CO_{2(g)} \rightarrow CaCO_{3(s)} + H_2O_{(l)}$–

If excess CO_2 is bubbled through, the cloudiness disappears (goes clear again) due to formation of soluble calcium hydrogencarbonate.

–$CaCO_{3(s)} + CO_{2(g)} + H_2O_{(l)} \rightarrow Ca(HCO_3)_{2(aq)}$–

2–Agriculture

Lime water is used to neutralize and control acidic soils to improve crop growth.

Acidic soils contain H^+ ions (from acids like nitric acid from fertilizers). The OH^- ions from lime water neutralize them: $H^+ + OH^- \rightarrow H_2O$

Uses of group 2 metals and thier compounds (continued)

Barium

1–Testing for sulfate

–$Ba^{2+}{}_{(aq)} + SO_4{}^{2-}{}_{(aq)} \rightarrow BaSO_{4(s)}$–

Note: HNO_3 is added before adding Barium nitrate (or chloride) solution to remove false positive results, especially those caused by $BaCO_3$ (also forms a white ppt.)

2–Barium meal

A drink containing Barium Sulfate which is insoluble in water (and in stomach acid).

Therefore, the Ba^{2+} ions cannot get into the bloodstream.

It is used in medical imaging as it shows up well on X–ray unlike soft tissues

(if a soluble Ba salt was used they would be poisonous)

Flame Test (for group 1 & 2)

Procedure:

- Clean the nichrome wire (cuz its unreactive) by dipping it in concentrated hydrochloric acid (HCl) and heating it in a Bunsen flame until no color is produced.
  –[This (HCl) removes any contaminants from previous tests]–
- Dip the clean wire into the solid sample (or solution) to pick up a small amount.
  –[HCl could be added to solution to form chloride compounds making the color clear]–
- Place the wire into the Bunsen flame (the blue part).
- Observe and record the color of the flame produced.

Explanation:

Energy absorption: When heated, electrons in the metal ions absorb energy and jump to a higher energy level (excited state).

Energy emission: The electrons are unstable in the higher level and fall back down to their original (ground) level.

Light released: As they fall, they release energy in the form of visible light.

Different colors: Each metal ion has a different energy gap between its electron shells.

A bigger energy gap → more energy released → shorter wavelength (towards blue/violet)

A smaller energy gap → less energy released → longer wavelength (towards red)

ion	Symbol	Flame Color
Lithum	Li^+	Red
Sodium	Na^+	Yellow / Orange
Potassium	K^+	Lilac
Rubidium	Rb^+	Red / Purple
Caesium	Cs^+	Blue / Violet
Beryllium	Be^{2+}	No color
Magnesium	Mg^{2+}	No color
Calcium	Ca^{2+}	(Brick) red
Strontium	Sr^{2+}	(Crimson) Red
Barium	Ba^{2+}	(Apple) green

Volumetric analysis

Other tests

Carbonate ions, CO^{2-}, and hydrogencarbonate ions, HCO^{-}
Test: Add dilute acid (HCl or HNO_3).
Observation: Effervescence (bubbles) of a colorless gas.
Confirming the gas is CO_2:
Pass the gas through lime water ($Ca(OH)_2$ solution).
 Positive result: Lime water turns cloudy / milky.

Testing for Ammonium Ions (NH_4)
Test: 1)Add sodium hydroxide solution (NaOH) to the substance being tested
 2)Warm the mixture gently.
Observation: Effervescence (bubbles) of a colorless gas with a pungent, choking smell.

$-NH_{4(aq)} + OH^-_{(aq)} \rightarrow NH_{3(g)} + H_2O_{(l)}-$ (Na^+ is spectator)

Confirming Ammonia Gas (NH_3):
–Method 1: Damp Red Litmus Paper
 Test: Hold a piece of damp red litmus paper near the mouth of the test tube.
 Observation: The damp red litmus paper turns blue. (Ammonia is an alkaline gas)

–Method 2: Concentrated HCl (Ammonia Fountain / White Fumes)
 Test: Dip a glass rod in concentrated hydrochloric acid (HCl) and hold it near the mouth of the test tube (where the gas is escaping).
 Observation: white smoke is produced.
 What happens: Thez ammonia gas reacts with hydrogen chloride gas (from the conc. HCl) to form tiny solid crystals of ammonium chloride (NH_4Cl) , which appear as white smoke.

$-NH_{3(g)} + HCl_{(g)} \rightarrow NH_4Cl_{(s)}-$

Testing for sulfate (already mentioned)
Test: Add barium nitrate($BaNO_3$) and nitric acid (HNO_3).
Observation: white precipitate.

$-Ba^{2+}{}_{(aq)} + SO_4{}^{2-}{}_{(aq)} \rightarrow BaSO_{4(s)}-$

Introduction

Volumetric analysis is a method used to determine the concentration of a unknown substance. It works by precisely measuring the volume of a solution of known concentration (standard solution) that reacts completely with a known volume of the solution you are analysing. The technique used to carry this out is called a titration.

Standard solution

A standard solution (volumetric solution) is a solution whose concentration is known with **high precision and accuracy** . It is the "known" solution you use in a titration to find the "unknown" concentration of another solution.

The standard solution is prepared by dissolving a known mass of a subtance (known as the primary standard) in a known volume of water → A solution with a know concentration.

To ensure high percision and accuray of standard solution the primary standard must:

- Solid
- High Purity: Impurities would lead to inaccurate concentrations.
- High Stability: It should not react with the air (e.g., absorb water or carbon dioxide) or decompose under normal conditions.
- High Molar Mass: minimises the percentage error when weighing the solid.
- Soluble in water
- Anhydrous: To avoid errors from water of crystallisation content changing.
- Must react completely and rapidly

Preparing a Standard Solution

- Calculate the mass of the primary standard needed for your desired concentration and volume, so if you want to prepare a 250 cm^3 of 0.1 mol dm^{-3} sulfamic acid:

–[$n = c \times V = 0.1 \times 0.25 = 0.025$ mol → $m = n \times M = 0.025 \times 97.1 = 2.4275$ g So it would about 2.4]–
(note: the mass used is not required to be exactly equal)

- Weigh the mass of solid accuratly and record exact mass
 –[Mass of bottle containing solid – mass of empty bottle (might have some traces)]–
- Transfer to a clean beaker and dissolve by ading about 100 cm^3 of distilled water and stir with a glass rod until all the solid has completely dissolved.
- Transfer solution into a 250 cm^3 volumetric flask using a funnel and then add washings. –[Washings are the result of rinsing the beaker, funnel and rod to ensure no solid is lost (all is transfered]–
- Add deionised water to the flask until you reach the graduation mark.
 –[A dropper would be used at the end to add drop of water to make up excatly to the mark]–
- Mix: Stopper the flask securely. Invert it and shake vigorously several times to ensure uniform concentration/homgeneous.

Volumetric analysis

Titration

Once you have your standard solution, you use it in a titration to find the concentration of an unknown solution

Method:

• Rinse Apparatus: Rinse the burette with deionised water then the standard solution. Rinse the pipette with with deionised water then the unknown solution. Rinse the conical flask with deionised water only.

> –Rinsing with water → removes impurities and other solutions
> –Risnsing with solution after water (standard solution for burette and unkown solution for pipette) → To remove water that would dilute the titran/analyt, changing its concentration. For the pipette, this ensures the number of moles delivered is exactly what you calculate.
> –Conical flask Rinsed with water only → Water dilutes the analyte but does not change the moles present. Rinsing with analyte would add extra moles.

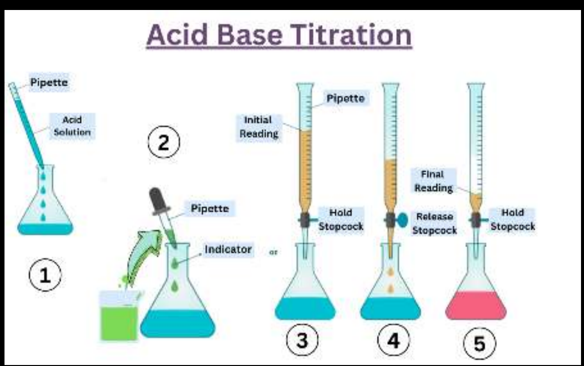
• Transfer Analyte: Use the pipette to transfer exactly 25.0 cm^3 (or the specified volume) of the unknown solution into the conical flask. Add a few drops of indicator (methyl orange or phenolphthalein).

• Fill the burette with the standard solution (Record initial burette reading, ensuring no air bubbles in the jet)

• Slowly add the standard solution to the conical flask, swirling constantly until the indicator just changes permanently (Record the final burette reading)

–[Near the end-point, add the solution dropwise]–

• Repeat for Accuracy: Carry out a 'rough' titration first to find the approximate end-point. Then, perform accurate titrations at least three times. Calculate and use the mean titre from your concordant results. (results within 0.1–0.2 cm^3 of each other)



Ensuring Acurracy (some of these are mentioned above):

1. Shaking/Inverting volumetic flask → Uniform concentration
2. Using a dropper to make up to the mark in the volumetric flask → accuratly measured
3. Rinsing measuring apparatus and adding washings → All solution is transfered
4. Reading from bottom of meniscus at eye level → accurate reading (avoiding parllax)
5. Ensure the jet space below the burette tap is filled with solution by running some of the solution down→ An air bubble released during titration will be recorded as titre, leading to an inaccurate reading.
6. Swirling flask → reactant are mixed thoroughly
7. Calculate the mean titre using only the concordant results → Discard any rough or anomalous results.

Indicator	In Acid	In Alkali	Endpoint Color (neutral)
Methyl orange	Red	Yellow	Orange
Phenolphthalein	Colorless	Pink	Pale

Rough titre

Purpose: To get an approximate idea of where the endpoint is.

Procedure:

• Add the titrant from the burette quickly (in 1–2 cm^3 increments) while swirling.

• Stop when the indicator just changes color (endpoint).

• Record this volume as the rough titre (e.g., 23.5 cm^3).

Why? So you know roughly how much to add in the accurate titrations, allowing you to add drop–by–drop near the end.

Accurate titre

Purpose: To get precise, reproducible results for calculation.

Procedure:

• Run the titrant quickly until you are within 1–2 cm^3 of the rough titre.

• Then add drop–by–drop until the indicator just changes color permanently.

• Record the final volume (to 2 decimal places, e.g., 23.45 cm^3).

Repeat: Do this until you have at least two concordant results (within 0.10 cm^3 of each other).

Causes for deviation from actual titre volume

Above Actual (Titre volume is higher than it should be)

1. Rough titration: The first titration is done quickly to find the approximate endpoint, so it is often overshot, giving a higher volume.

2. Overshooting endpoint: Adding titrant too fast or not swirling enough means you pass the endpoint before realizing, resulting in a higher volume.

3. Burette rinsed with water only: Leftover water in the burette dilutes the titrant. A weaker titrant means more volume is needed to neutralize the analyte, giving a higher titre.

4. Burette jet space filled with air bubble: If the space below the tap is not filled with titrant before the initial reading, the bubble occupies that volume. During titration, the bubble escapes and is replaced by titrant. This extra titrant is included in the final volume, giving a higher titre reading.

Below Actual (Titre volume is lower than it should be)

1. Pipette rinsed with water only: Leftover water dilutes the analyte, meaning fewer moles of acid or base are present. Less titrant is needed to react, giving a lower titre

Factor the affect both

1. Parallax error: reading from above → gives a volume below actual
reading from below → gives a volume above actual

2. Incorrect dilution: caused by adding more or less water(solvent) to the analyte or titrant, and might be caused by incorrect rinsing method (mentioned above)

> If titrant is too dilute (e.g., water left in burette) → higher titre.
> if titrant is too concentrated (e.g., evaporation or preparation error) → lower titre.
> if analyte is too dilute (too much solvent added) → fewer moles in pipette → lower titre.
> If analyte is too concentrated (too little solvent added in flask) → more moles in pipetted volume → higher titre.

Group 7 (Halogens)

(limited to chlorine, bromine and iodine)

Properties

General
- They exist as diatomic molecules
- Chlorine and bromine are poisonous

As you go down the group
- Atomic radius increases
- Density increases

—• Melting and boiling temperatures increase

Reason: Halogens exist as simple covalent molecules held by london forces, so down the group number of electrons in the molecule increases → electron cloud becomes larger and more distortable → creates stronger induced dipole—dipole forces between the molecules → More energy is required to overcome these stronger force.

—• Electronegativity decreases

Electronegativity: The ability of an atom to attract the bonding electrons in a covalent bond.

Reason: atomic radius increases + increasing number of shells (more shielding) + distance between nucleus and electron increases → outweigh increase in nuclear charge → attraction between the nucleus and the bonding pair of electrons becomes weaker → ***electronegativity decreases***

—• Reactivity decreases

Reason: Halogens are non—metals that react by gaining one electron (oxidizing agent), so down the group the electronegativity decreases (mentioned ↑) → Nuclear attraction to incoming electrons decreases → less oxidizing power → harder to gain electrones → ***less reactive***

—• Their color becomes darker

Halogen	Color at Standard Conditions (RTP)	Color in Aqueous Solution	Color in a hydrocarbon Solvent (e.g. cyclohexane)
Chlorine (Cl_2)	Pale green gas	Pale green / colourless	Pale green
Bromine (Br_2)	Red-brown liquid (gives off red-brown vapour)	Yellow / Orange	Yellow / Orange
Iodine (I_2)	Grey-black shiny solid(sublimes into Purple gas)	Brown (fades to yellow)	Purple

Reaction of halogens

Oxidation reactions with Group 1 & 2 metals (redox reactions)

Halogens and metal react forming metal hlids salts, where:

Halogens act as oxidising agents (they gain electrons, their oxidation number decreases from 0 to —1).

Metals act as reducing agents (they lose electrons, their oxidation number increases).

•—Vigorousness—•

The most vigorous reaction hapeens between the bottom of group 1—2 and top of group 7

With group 1: $2M+X_2 \rightarrow 2MX$

With group 2: $M+X_2 \rightarrow MX_2$

Disproportionation: Chlorine with Water

Reaction: Chlorine reacts with water to form hydrochloric acid and chloric(I) acid (hypochlorous acid).

disproportionation reaction: reaction in which an element in a single species is being simultaneously oxidized and reduced

Note: this reaction is reversable so (toxic) Cl_2 would be produces from the reaction of HCl with HCLO

reduced: 0 → -1

$$Cl_{2(aq)} + H_2O_{(l)} \rightarrow HCl_{(aq)} + HClO_{(aq)}$$

oxidized: 0 → +1

HClO (chloric(I) acid) is a powerful oxidising agent.

hence it used in water treatment to kills bacteria.

It also oxidises harmful coloured organic compounds, bleaching them colourless.

Disproportionation: Chlorine with alkali

1. Cold alkali (15°c)

reduced: 0 → -1

$$Cl_{2(aq)} + 2NaOH_{(aq)} \rightarrow NaCl_{(aq)} + NaClO_{(aq)} + H_2O_{(l)}$$

oxidized: 0 → +1

Product: Sodium chlorate(I) [NaClO] is a bleach

Use: used in household bleach and to disinfect water

2. Hot alkali (70°c)

Sodium chlorate (V)

reduced: 0 → -1

$$3Cl_{2(aq)} + 6NaOH_{(aq)} \rightarrow 5NaCl_{(aq)} + NaClO_{3(aq)} + H_2O_{(l)}$$

oxidized: 0 → +5

Why does hot alkali give chlorate(V)?

Cold: First forms chlorate(I) (bleach).

Hot: Chlorate(I) is unstable when heated. It further disproportions to form chlorate(V) and more chloride.

$$3ClO^- \rightarrow 2Cl^- + ClO_3^-$$

Trend down the group:

This second step happens more easily as you go down.

Iodine forms iodate(V) so easily it happens even in cold conditions.

Displacement Reactions (Halogen + Halide)

Rule: A more reactive halogen displaces a less reactive halogen from its halide solution.

Trend: Reactivity decreases down the group (Fluorine > Chlorine > Bromine > Iodine).

Displacement?	$KCl_{(aq)}$	$KBr_{(aq)}$	$KI_{(aq)}$
Cl_2	—	Displaces (Yellow) $Cl_2 + 2Br^- \rightarrow Br_2 + 2Cl^-$	Displaces (Brown) $Cl_2 + 2I^- \rightarrow I_2 + 2Cl^-$
Br_2	No Reaction	—	Displaces (Brown) $Br_2 + 2I^- \rightarrow I_2 + 2Br^-$
I_2	No Reaction	No Reaction	—

Group 7 (Halogens)

Halides vs Halogens

Halogens (X_2): Act as oxidising agents. They gain electrons to form halide ions.

$$X_2 + 2e^- \rightarrow 2X^-$$

Halides (X^-): Act as reducing agents. They lose electrons to form halogens.

$$2X^- \rightarrow X_2 + 2e^-$$

Trend:

OXIDISING POWER	HALOGEN	HALIDE	REDUCING POWER
High	fluorine F_2	fluoride F^-	Low
↓	chlorine Cl_2	chloride Cl^-	↓
	bromine Br_2	bromide Br^-	
	iodine I_2	iodide I^-	
Low	astatine At_2	astatide At^-	High

This is because:

Down the group, halide ions get larger.

The outer electron is further from the nucleus and more shielded.

So the nuclear pull on the outer electron decreases

It is easier to lose an electron → better reducing agent.

Reaction of halides

Reaction with concentrated sulfuric acid

What happens: Concentrated H_2SO_4 by acting as an acid reacts with solid Group 1 halides to produce hydrogen halides (HX).

If the metal has a strong enough reducing power: Concnetrated H_2SO_4 will act as an oxidising agent and the extent of its reducing depend on how strong of a reducing agent the halid is

1. Chloride (Cl^-) -- Weak reducing agent (not redox)

$$NaCl_{(s)} + H_2SO_{4(l)} \rightarrow NaHSO_{4(s)} + HCl_{(g)}$$

2. Bromide (Br^-) -- Stronger reducing agent

Stage 1 (Same as chloride):

$$NaBr_{(s)} + H_2SO_{4(l)} \rightarrow NaHSO_{4(s)} + HBr_{(g)}$$

Stage 2 (Redox occurs): HBr reduces some of the H_2SO_4.

$-1 \xrightarrow{\text{oxidized}} 0$

$$2HBr_{(g)} + H_2SO_{4(l)} \rightarrow Br_{2(g)} + SO_{2(g)} + 2H_2O$$

$+6 \xrightarrow[\text{reduced}]{} +4$

Iodide (I^-) -- Strongest reducing agent

Stage 1 (Same as others):

$$NaI_{(s)} + H_2SO_{4(l)} \rightarrow NaHSO_{4(s)} + HI_{(g)}$$

Stage 2 (More vigorous redox): HI is such a strong reducing agent it reduces H_2SO_4 further, often to multiple products.

$-1 \xrightarrow{\text{oxidized}} 0$

$$2HI_{(g)} + H_2SO_{4(l)} \rightarrow I_{2(g)} + SO_{2(g)} + 2H_2O_{(l)}$$

$+6 \xrightarrow[\text{reduced}]{} +4$

$-1 \xrightarrow{\text{oxidized}} 0$

$$6HI_{(g)} + H_2SO_{4(l)} \rightarrow 3I_{2(g)} + S_{(s)} + 4H_2O_{(l)}$$

$+6 \xrightarrow[\text{reduced}]{} 0$

$-1 \xrightarrow{\text{oxidized}} 0$

$$8HI_{(g)} + H_2SO_{4(l)} \rightarrow 4I_{2(g)} + H_2S_{(g)} + 4H_2O_{(l)}$$

$+6 \xrightarrow[\text{reduced}]{} -2$

HALIDE	OBSERVATIONS	PRODUCTS
NaCl	• **Misty/steamy white fumes**	• Hydrogen chloride - HCl
NaBr	• **Misty/steamy fumes** • **Brown fumes** • Colourless gas with **choking smell**	• Hydrogen bromide - HBr • Bromine - Br_2 • Sulfur dioxide - SO_2
NaI	• **Misty/steamy fumes** • **Purple** vapor/ black solid • Colourless gas with **choking smell** • **Yellow solid** deposits • Colourless gas with **rotten egg smell**	• Hydrogen iodide - HI • Iodine - I_2 • Sulfur dioxide - SO_2 • Sulfur - S • Hydrogen sulfide - H_2S

Why sulfuric acid is unsuitable for HBr/HI production:

Sulfuric acid has a strong enough oxidising power to oxidise bromide and iodide. That is why it is not a suitable acid to produce pure hydrogen bromide or hydrogen iodide. Phosphoric acid (H_3PO_4) could be used instead as it is a weaker oxidising agent.

Hydrogen Halids reaction with Ammonia Gas (NH_3)

What happens: Acid–base reaction. Hydrogen halide (acid) reacts with ammonia (base) to form a white ammonium halide salt.

Observation: Dense white smoke.

General equation:

$$NH_{3(g)} + HX_{(g)} \rightarrow NH_4X_{(s)}$$

Hydrogen Halids reaction with water

What happens: Hydrogen halides dissolve and react with water to form acidic solutions. They donate a proton (H^+) to water.

Observation: The gas dissolves rapidly, often creating a mist/fumes. The solution turns blue litmus red.

General equation:

$$HX + H_2O \rightarrow H_3O^+ + X^-$$

Note: for hydorgen fluoride a reversable arrow "⇌" is used, as they form weak acids

Testing for Halides

Reaction with Sulfuric acid

(details already mentioned)

Reaction with Silver Nitrate (in presence of nitric acid)

What happens: Silver ions (Ag^+) react with halide ions (X^-) to form a silver halide precipitate.

Role of nitric acid: Added first to remove any interfering ions (like carbonate, phosphate) that also form precipitates with Ag^+. It ensures any precipitate is only from a halide.

Solubility in Aqueous Ammonia (NH_3)

This distinguishes between the halides, as they have different solubilities in ammonia

Halide Ion	Precipitate Colour	Effect of Dilute NH_3	Effect of Conc. NH_3
Cl^-	White - (AgCl)	Dissolves	Dissolves
Br^-	Cream - (AgBr)	No change	Dissolves
I^-	Yellow - (AgI)	No change	No change